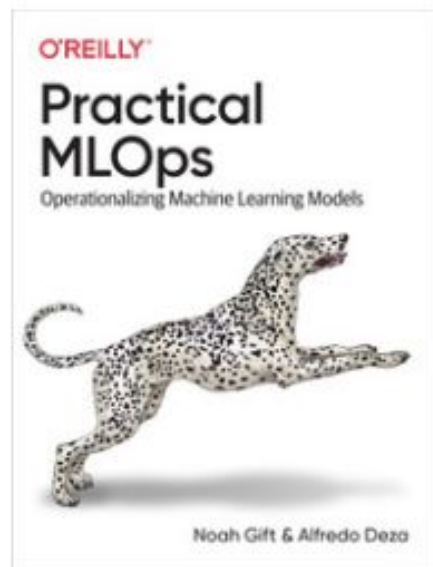


What is MLOps anyways



Practical MLOps

Noah Gift, Alfredo Deza

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Media, Inc.

Problem: Models dont make
it to production

History Context: Kaizen



The reason models are not moving into production is the impetus for the emergence of MLOps as a critical industry standard. MLOps shares a lineage with DevOps in that DevOps philosophically demands automation. A common expression is if it is not automated, it's broken. Similarly, with MLOps, there must not be components of the system that have humans as the levers of the machine. The history of automation shows that humans are the least valuable doing repetitive tasks but are the most valuable using technology as the architects and practitioners. Likewise, coordination between developers, models, and operations must coordinate through transparent teamwork and healthy collaboration. **Think of MLOps as the process of automating machine learning using DevOps methodologies.**

TL;DR = Data Science + DevOps

CLOUD NATIVE ML PLATFORMS



AWS
SageMaker



Azure
ML Studio



GCP
AI Platform

CONTAINERIZED WORKFLOWS



Docker
format
containers



Kubernetes



Private and
public
container
registries

SERVERLESS TECHNOLOGY



AWS
Lambda



AWS
Athena



Google
Cloud
Functions

BIG DATA PLATFORMS AND TOOLS



Databricks



Hadoop/Spark



Snowflake

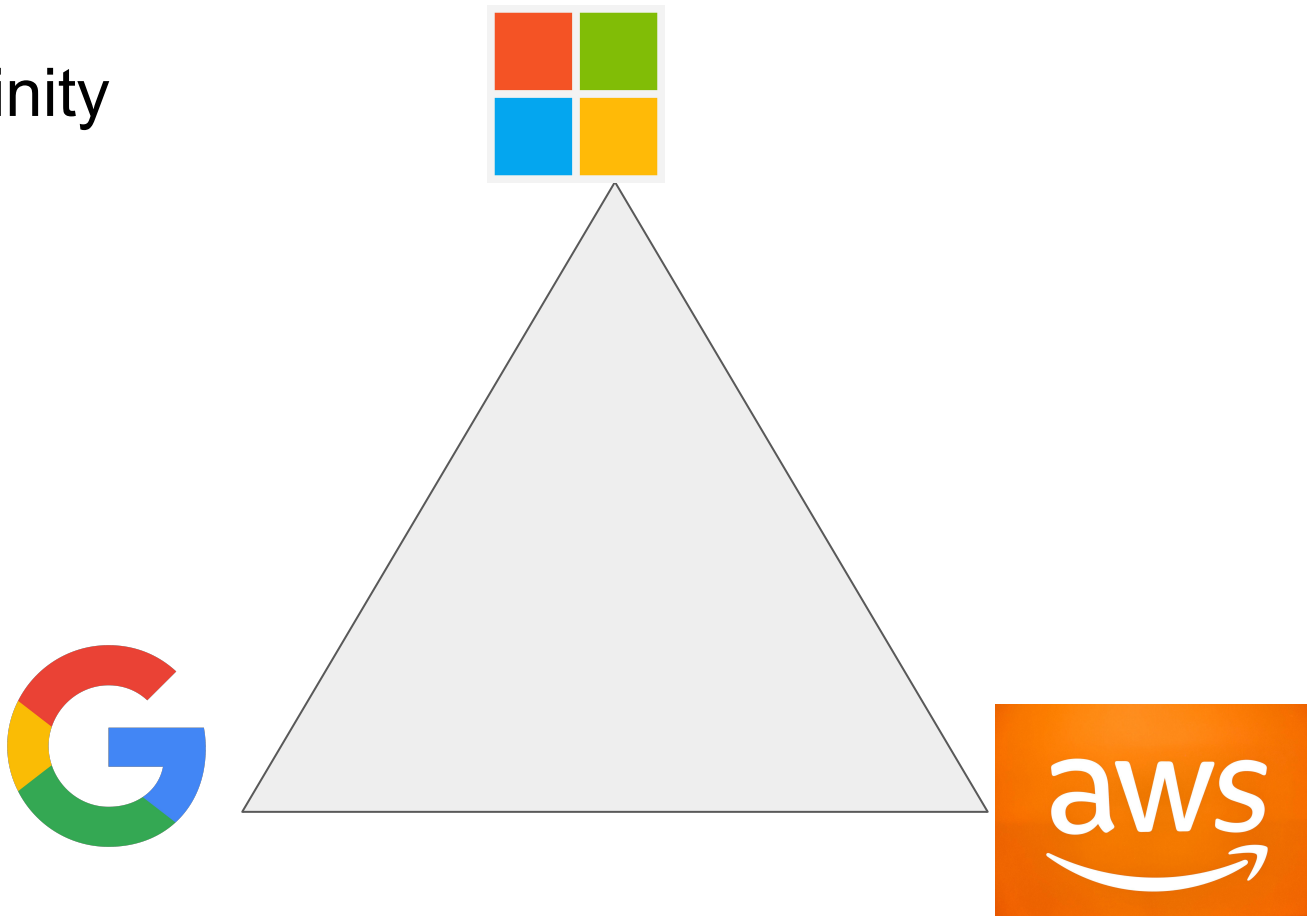


Amazon EMR



Google Big Query

The Trinity





CERTIFICATION

Microsoft Certified: Azure Data Scientist Associate

Manage data ingestion and preparation, model training and deployment, and machine learning solution monitoring with Python, Azure Machine Learning and MLflow.



AWS Certified Machine Learning - Specialty

AWS Certified Machine Learning - Specialty validates your expertise in building and deploying machine learning solutions in the AWS Cloud. This credential demonstrates to employers that you can architect ML/deep learning workloads, optimize model training, and implement production-ready ML systems following AWS best practices.

[Schedule your exam](#)



Professional Machine Learning Engineer

Certification exam guide

Why should you care?

Cloud	191K
AWS	83K
Azure	51K
GCP	21K
Data engineer	70K
Machine learning engineer	55K
<u>Data scientist</u>	20K

LinkedIn Job openings 2020



Continuous Integration

CI is the process of continuously testing a software project and improving the quality based on these tests' results. It is automated testing using open source and SaaS build servers such as GitHub Actions, Jenkins, Gitlab, CircleCI, or cloud native build systems like AWS Code Build.

Continuous Delivery

This method delivers code to a new environment without human intervention. CD is the process of deploying code automatically, often through the use of IaC.

Microservice

A microservice is a software service with a distinct function that had little to no dependencies. One of the most popular Python-based microservice frameworks is Flask. For example, a machine learning prediction endpoint is an excellent fit for a microservice. These microservices can use a wide variety of technologies, including FaaS (function as a service). A perfect example of a cloud function is AWS Lambda. A microservice could be container-ready and use CaaS (container as a service) to deploy a Flask application with a Dockerfile to a service like AWS Fargate, Google Cloud Run, or Azure App Services.

Infrastructure as Code

Infrastructure as Code (IaC) is the process of checking the infrastructure into a source code repository and “deploying” it to push changes to that repository. IaC allows for idempotent behavior and ensures the infrastructure doesn’t require humans to build it out. A cloud environment defined purely in code and checked into a source control repository is a good example use case. Popular technologies include cloud-specific IaC like AWS Cloud Formation or AWS SAM (Serverless Application Model). Multicloud options include Pulumi and Terraform.

Monitoring and instrumentation

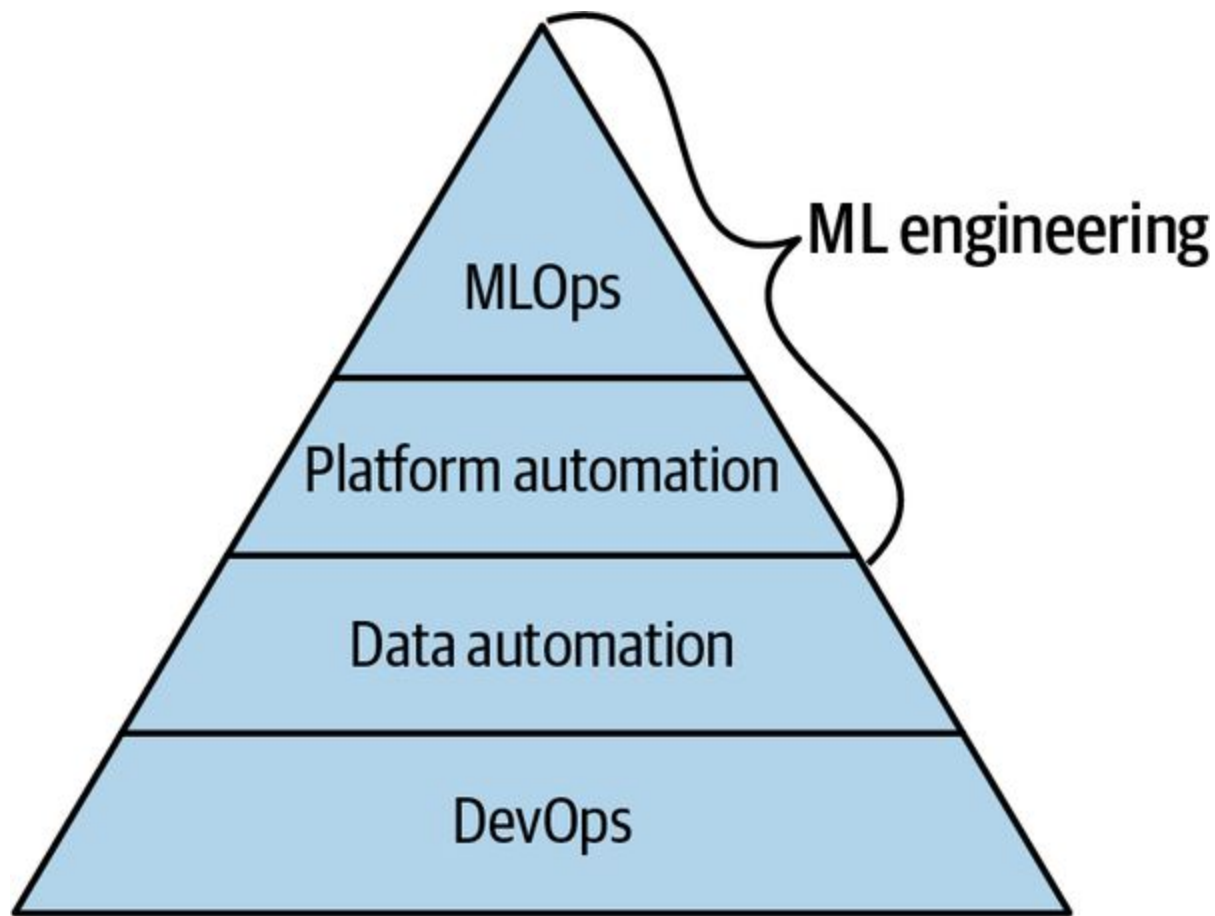
Monitoring and instrumentation are the processes and techniques used that allow an organization to make decisions about a software system's performance and reliability. Through logging and other tools like application performance monitoring tools such as New Relic, Data Dog, or Stackdriver, monitoring and instrumentation are essentially collecting data about the behavior of an application in production or data science for deployed software systems. This process is where Kaizen comes into play; the data-driven organization uses this instrumentation to make things better daily or weekly.

Effective technical communication

This skill involves the ability to create effective, repeatable, and efficient communication methods. An excellent example of effective technical communication could be adopting AutoML for the initial prototyping of a system. Of course, ultimately, the AutoML model may be kept or discarded. Nevertheless, automation can serve as an informational tool to prevent work on an intractable problem.

Effective technical project management

This process can efficiently use human and technology solutions, like ticket systems and spreadsheets, to manage projects. Also, appropriate technical project management requires breaking down problems into small, discrete chunks of work, so incremental progress occurs. An antipattern in machine learning is often when a team works on one production machine model that solves a problem “perfectly.” Instead, smaller wins delivered daily or weekly is a more scalable and prudent approach to model building.



ML engineering hierarchy of needs

Code: Make

Make install

This step installs software via the `make install` command

Make lint

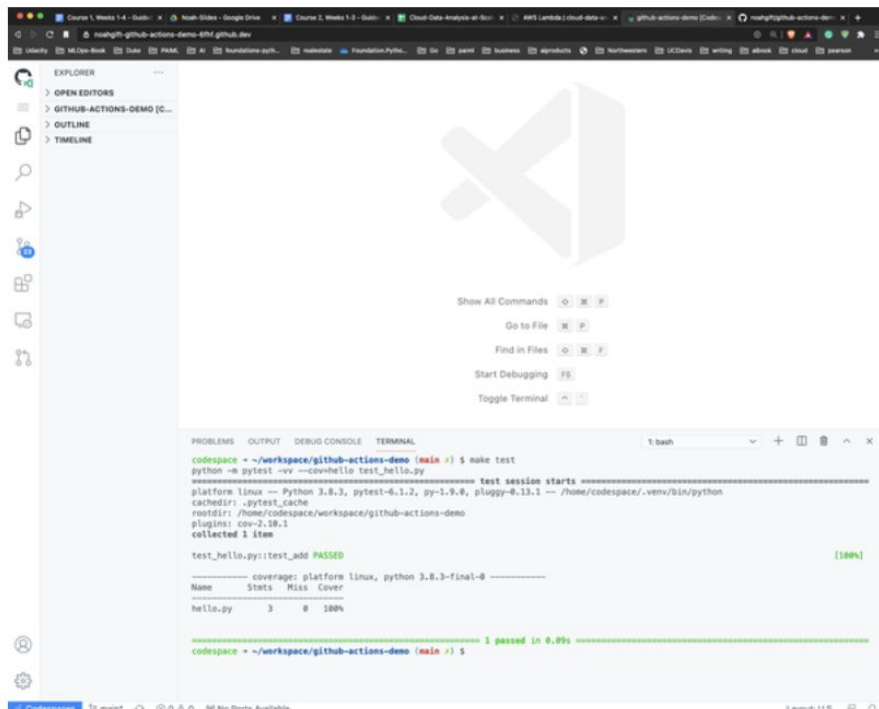
This step checks for syntax errors via the `make lint` command

Make test

This step runs tests via the `make test` command:

```
install:
    pip install --upgrade pip &&\
        pip install -r requirements.txt
lint:
    pylint --disable=R,C hello.py
test:
    python -m pytest -vv --cov=hello test_hello.py
```

```
$ make install
pip install --upgrade pip &&\
    pip install -r requirements.txt
Collecting pip
  Using cached pip-20.2.4-py2.py3-none-any.whl (1.5 MB)
[.....more output suppressed here.....]
```



```
$ make lint  
pylint --disable=R,C hello.py
```

```
-----  
Your code has been rated at 10.00/10
```

```
$ make test
python -m pytest -vv --cov=hello test_hello.py
===== test session starts =====
platform linux -- Python 3.8.3, pytest-6.1.2, \
/home/codespace/.venv/bin/python
cachedir: .pytest_cache
rootdir: /home/codespace/workspace/github-actions-demo
plugins: cov-2.10.1
collected 1 item

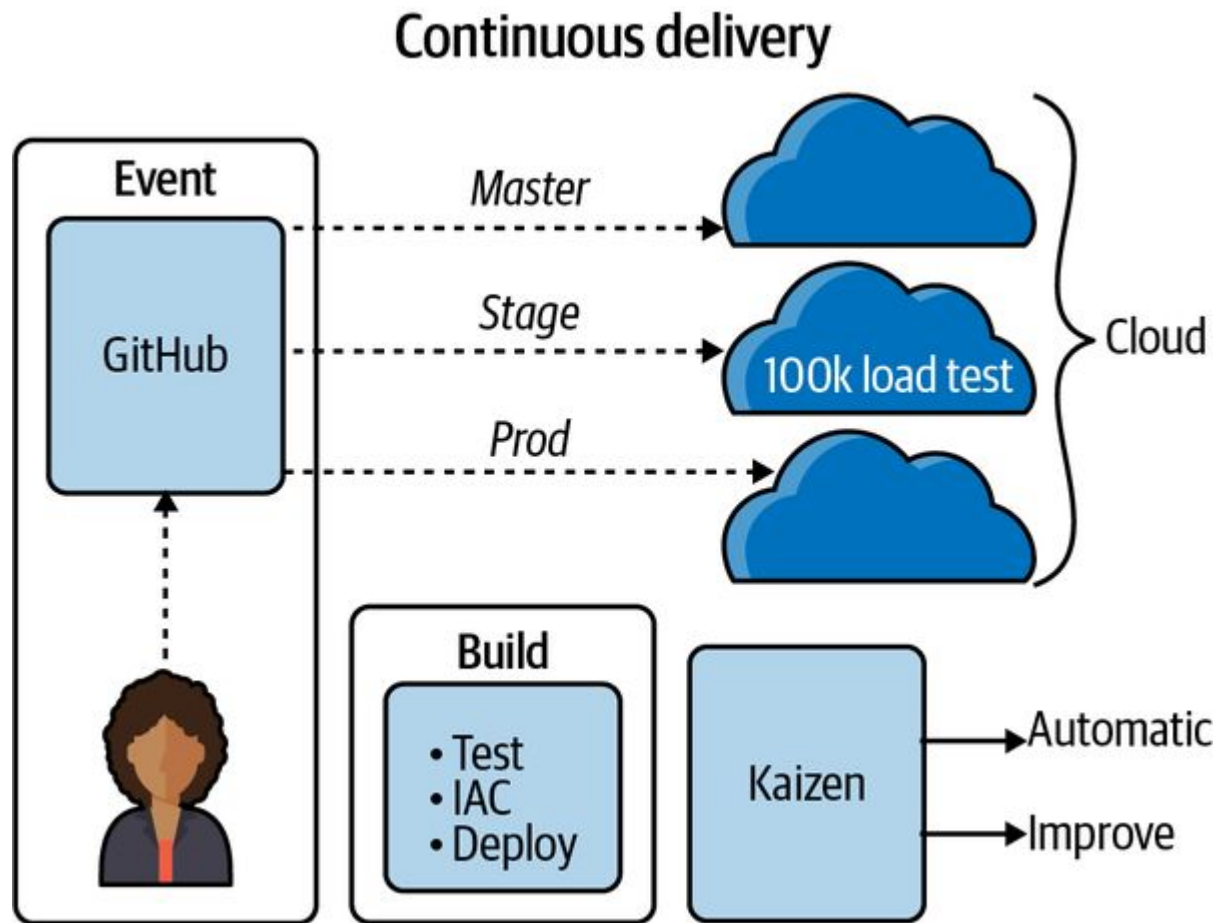
test_hello.py::test_add PASSED

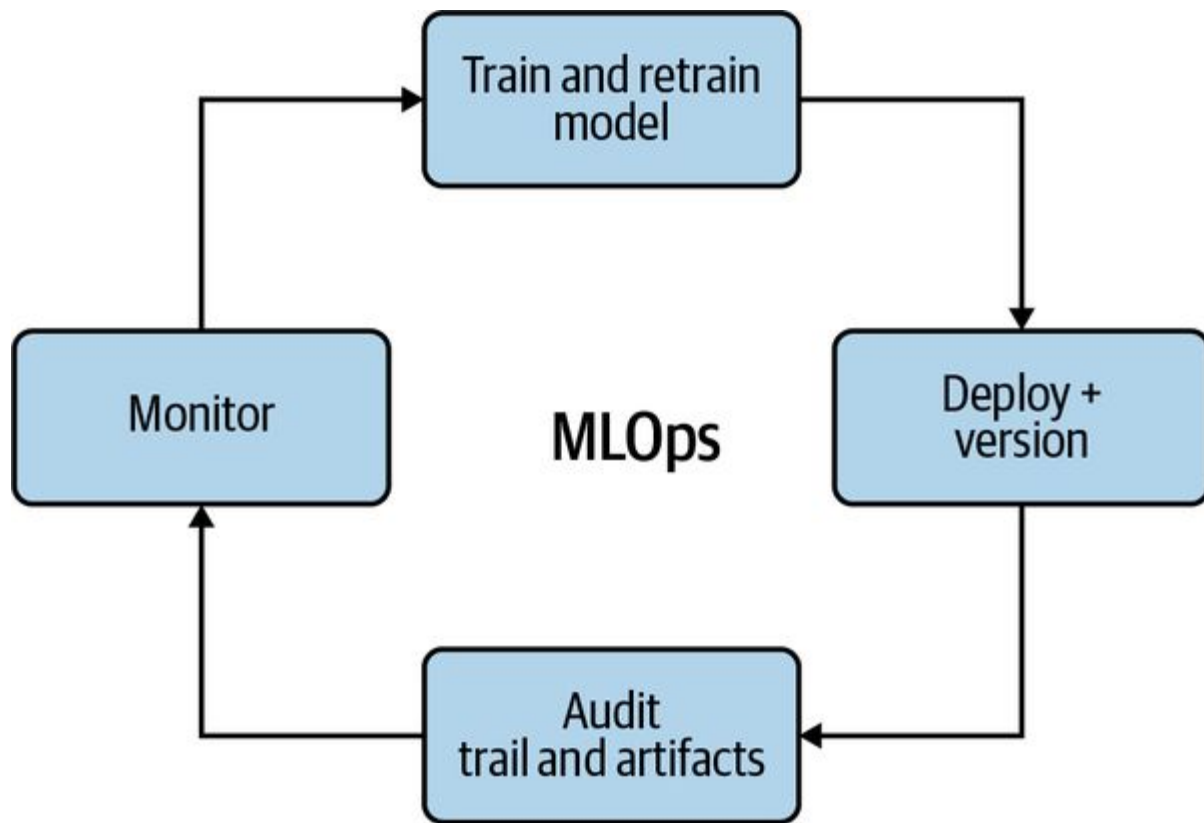
----- coverage: platform linux, python 3.8.3-final-0 -----
Name          Stmts    Miss  Cover
-----
hello.py         3         0   100%
```


Github Actions

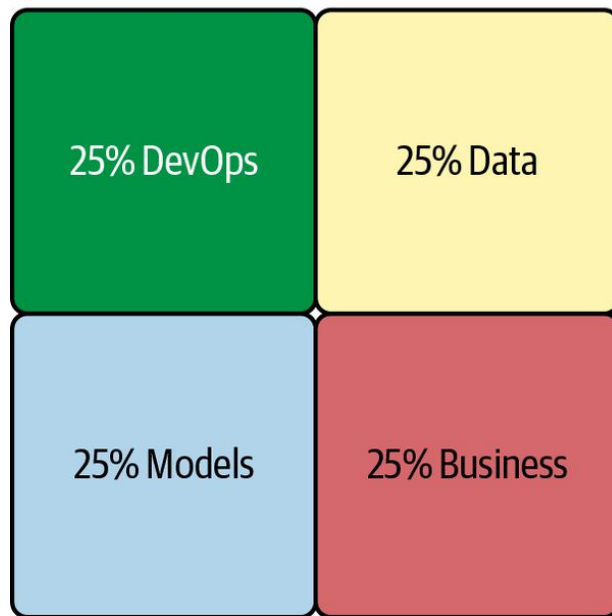
```
name: Azure Python 3.5
on: [push]
jobs:
  build:
    runs-on: ubuntu-latest
    steps:
      - uses: actions/checkout@v2
      - name: Set up Python 3.5.10
        uses: actions/setup-python@v1
        with:
          python-version: 3.5.10
      - name: Install dependencies
        run: |
          make install
      - name: Lint
        run: |
          make lint
      - name: Test
        run: |
          make test
```

CD: Git





MLOPS



Projects

Project Ideas

Example 1: SwipeFlix



- Product Offering: Replaces netflix doom scrolling, better UI/UX
- Is there Data available?
 - Yes! Famous [IMDB dataset](#)
 - Scrape RottenTomatoes
- Is there a usecase for ML?
 - Yes collaborative filtering
 - sentiment analysis
- Is there a usecase for LLM?
 - Yes! Majority Reviews are text
 - Movie descriptions titles are text

Example 2: Fake News Buster



- Product Offering: Find out Real vs Fake News
- Is there Data available?
 - Yes!
 - [AG News](#)
 - [Kaggle](#)
- Usecases for ML: Text classification with Naive Bayes, SVM, Logistic Regression.
- Usecases for LLM: Fine-tune model for news credibility detection, generate fact-checking summaries.

Example 3: Purrdict



Adobe Stock | #186197649

- Product Offering: Product Reviews TL;DR
- Is there Data available?
 - Yes!
 - [SNAP: Web data: Amazon reviews](#)
- Usecases for ML: Sentiment Analysis
-
- Usecases for LLM: Generate short reviews , Pros: Cons: ..